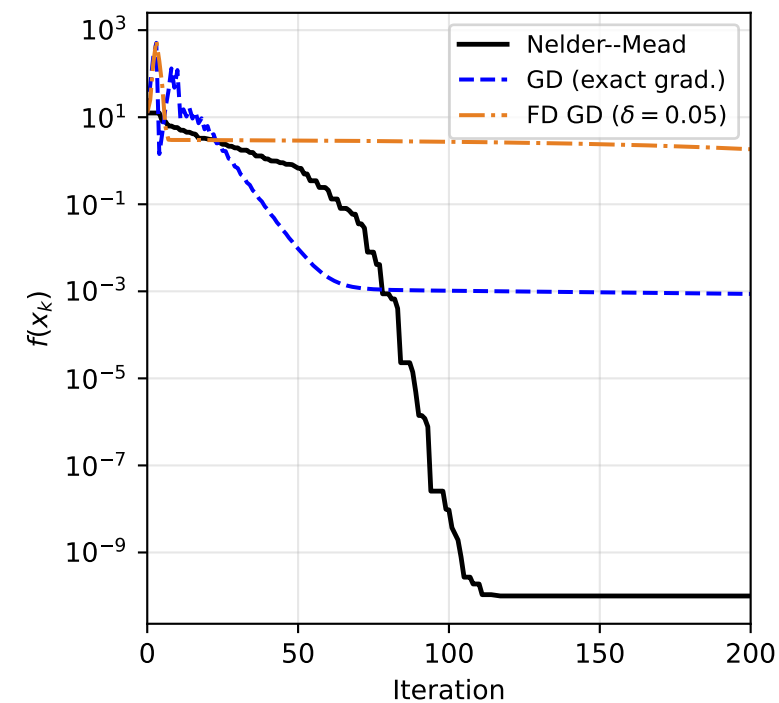
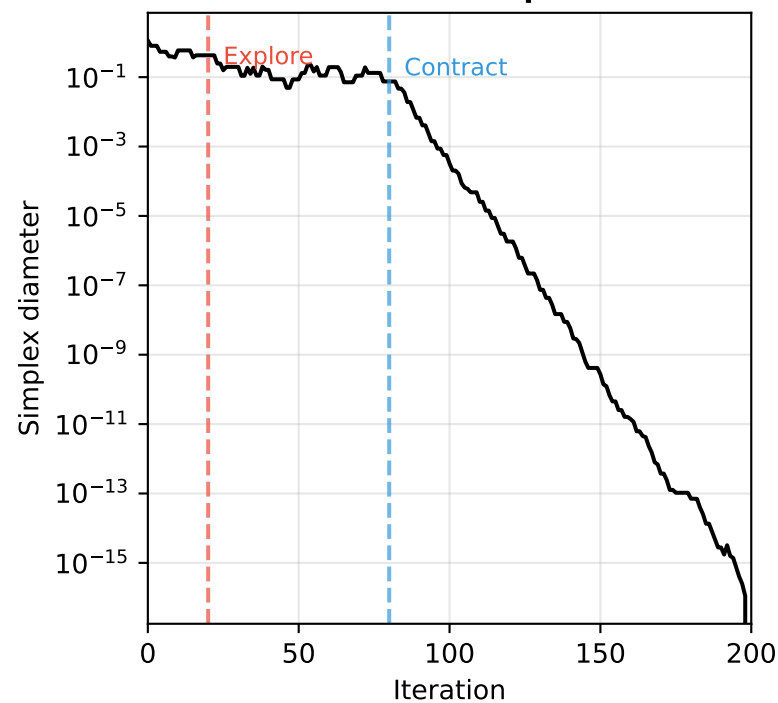


Q4: Derivative-Free Methods — Nelder-Mead Detail, FD Accuracy, Dimension Scaling

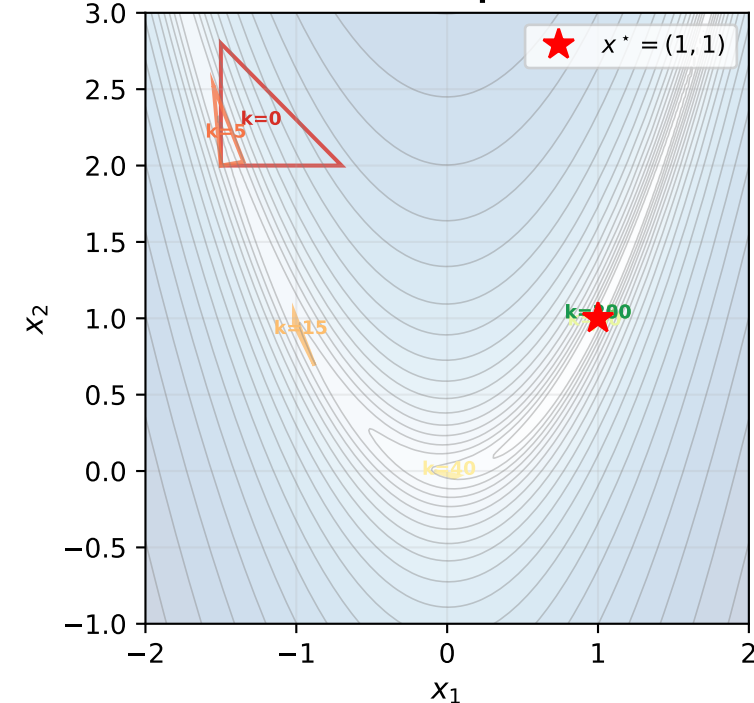
Nelder-Mead vs GD on Rosenbrock



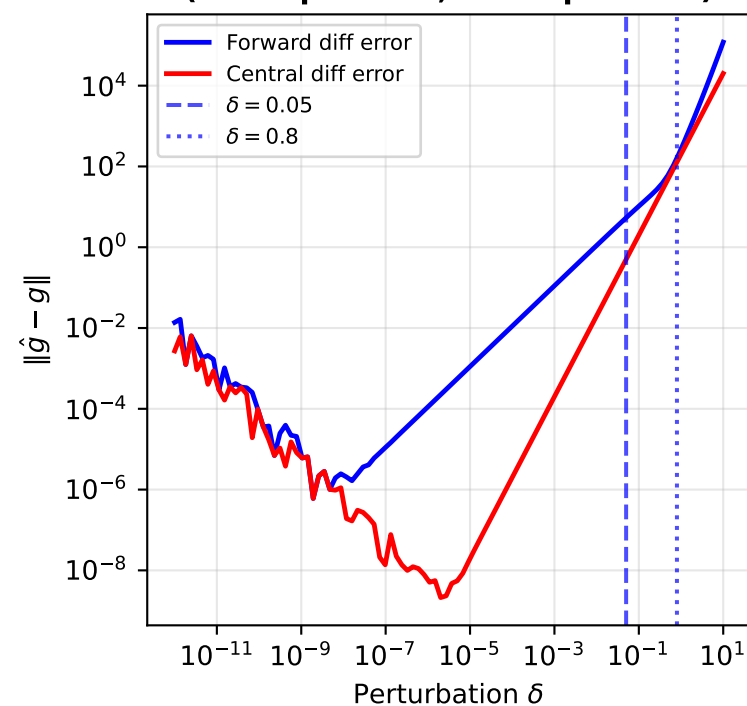
Nelder-Mead Simplex Size



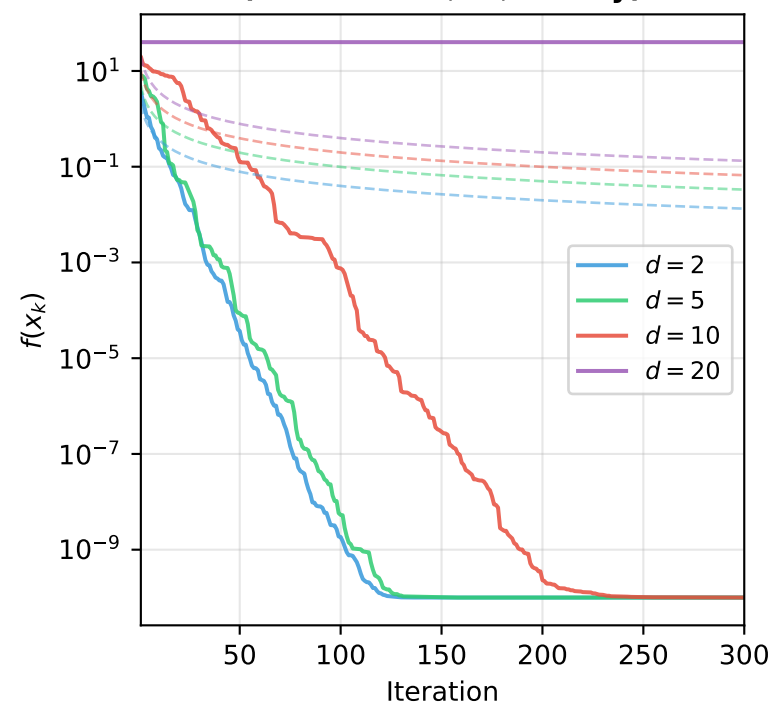
Nelder-Mead Simplex Evolution



FD vs CD: Gradient Error (FD slope=1.00, CD slope=2.00)



Random Search: $O(d/k)$ Rate vs Dimension (dashed = $O(d/k)$ theory)



Convergence vs Oracle Budget

